TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	92.2 ± 0.87
Correlation Stability	93.8 ± 0.75
Distribution Consistency	93.2 ± 0.87
Entropy Preservation	96.2 ± 0.40
Mean Z Score Drift	0.1 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	99.5 ± 0.50
Feature Accuracy	95.2 ± 0.75
Novelty Score	0.8 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: heart_failure_clinical_records.csv

1. Dataset Portrait

Attribute	Value
Rows	299
Columns	13
Numeric features	13
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	60.834	60.158	-0.676	1.1%	OK
anaemia	0.431	0.452	+0.020	4.7%	OK
creatinine_phosphokinase	581.839	510.753	-71.086	12.2%	Warn
diabetes	0.418	0.448	+0.030	7.1%	OK
ejection_fraction	38.084	39.293	+1.209	3.2%	OK
high_blood_pressure	0.351	0.368	+0.017	4.8%	OK
platelets	263358.029	258308.523	-5049.506	1.9%	OK
serum_creatinine	1.394	1.409	+0.015	1.1%	OK
serum_sodium	136.625	136.828	+0.203	0.1%	OK
sex	0.649	0.636	-0.013	2.0%	OK
smoking	0.321	0.310	-0.011	3.6%	OK
time	130.261	134.845	+4.584	3.5%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	92.2 ± 0.87
Correlation Stability (*2)	93.8 ± 0.75
Distribution Consistency (*3)	93.2 ± 0.87
Entropy Preservation (*4)	96.2 ± 0.40
Mean Z Score Drift (*5)	0.1 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	99.5 ± 0.50
Feature Accuracy (*10)	95.2 ± 0.75
Novelty Score (*11)	0.8 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.835	0.74	0.7866666666666666	-0.04833333333333332	*
F1	0.7547418227770806	0.5802690718668546	0.6782631310418318	-0.07647869173524882	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8450000000000001	0.7716666666666667	0.825	-0.020000000000000007	
F1	0.7616747329247329	0.5944508010975769	0.7248054184151819	-0.03686931450955102	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7499999999999999	0.7483333333333333	0.7483333333333333	-0.0016666666666666718	
F1	0.5667171896628354	0.588346517669504	0.575092016433168	0.00837482677033265	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6133333333333335	0.605	0.6316666666666666	0.018333333333333313	
F1	0.007142857142857143	0.10376589266041598	0.12749052396878482	0.12034766682592771	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6333333333333334	0.6333333333333334	0.6333333333333334	0	
F1	N/A	0.0	N/A	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

ACCURACY	0.825	0.7716666666666667	0.8150000000000001	-0.01000000000000002	
F1	0.7417404771442427	0.6429619395677512	0.7219798530284852	-0.0197606241157575	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8283333333333334	0.7499999999999999	0.8099999999999999	-0.01833333333333323	
F1	0.7517499705786448	0.5858885336351254	0.711676319003508	-0.04007365157513685	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.4866666666666664	0.5216666666666667	0.6083333333333334	0.12166666666666667	
F1	0.2513438077867478	0.2273574619967166	0.1449241326725124	-0.10641967511423547	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.78	0.6766666666666666	0.7733333333333333	-0.00666666666666654	
F1	0.6671367569667771	0.5046827031105615	0.6640883147404887	-0.00304844222628395	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8066666666666666	0.7833333333333333	0.7983333333333333	-0.00833333333333314	
F1	0.6937863401621888	0.6391451954714027	0.6740892136832769	-0.01969712647891975	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
2	55.0	0.0	7861.0	0.0	38.0	0.0	263358.03	1.1	136.0	1.0
3	65.0	0.0	146.0	0.0	20.0	0.0	162000.0	1.3	129.0	1.0
4	50.0	1.0	111.0	0.0	20.0	0.0	210000.0	1.9	137.0	1.0
5	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
6	90.0	1.0	47.0	0.0	40.0	1.0	204000.0	2.1	132.0	1.0
7	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0
8	60.0	1.0	315.0	1.0	60.0	0.0	454000.0	1.1	131.0	1.0
9	65.0	0.0	157.0	0.0	65.0	0.0	263358.03	1.5	138.0	0.0
10	80.0	1.0	123.0	0.0	35.0	1.0	388000.0	9.4	133.0	1.0
11	75.0	1.0	81.0	0.0	38.0	1.0	368000.0	4.0	131.0	1.0
12	62.0	0.0	231.0	0.0	25.0	1.0	253000.0	0.9	140.0	1.0
13	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
14	50.0	1.0	168.0	0.0	38.0	1.0	276000.0	1.1	137.0	1.0
15	49.0	1.0	80.0	0.0	30.0	1.0	427000.0	1.0	138.0	0.0
16	82.0	1.0	379.0	0.0	50.0	0.0	47000.0	1.3	136.0	1.0
17	87.0	1.0	149.0	0.0	38.0	0.0	262000.0	0.9	140.0	1.0
18	45.0	0.0	582.0	0.0	14.0	0.0	166000.0	0.8	127.0	1.0
19	70.0	1.0	125.0	0.0	25.0	1.0	237000.0	1.0	140.0	0.0
20	48.0	1.0	582.0	1.0	55.0	0.0	87000.0	1.9	121.0	0.0

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	57.0	1.0	60.0	0.0	40.0	0.0	235333.0	0.8	133.0	0.0
2	63.0	0.0	177.0	0.0	35.0	0.0	220000.0	1.8	139.0	0.0
3	48.1329981838054 34	0.0	250.0	0.0	60.0	0.0	232655.0	3.44246722917033 58	141.0	0.0
4	55.0	0.0	170.0	0.0	36.0	1.0	219099.0	0.9	136.0	1.0
5	75.0	0.0	224.0	0.0	38.0	1.0	203219.0	3.0	140.0	0.0
6	68.0	0.0	5087.0	0.0	50.0	1.0	283078.0	0.9	138.0	0.0
7	45.0	1.0	66.0	1.0	25.0	0.0	233000.0	0.8	135.0	1.0
8	70.0	0.0	161.0	0.0	25.0	0.0	244000.0	1.2	142.0	0.0
9	58.0	1.0	400.0	0.0	40.0	0.0	164000.0	1.0	139.0	0.0
10	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	55.0	0.0	80.0	0.0	40.0	0.0	266345.0	1.0	136.0	1.0
2	51.0	0.0	299.0	1.0	40.0	0.0	368000.0	1.0	140.0	1.0
3	63.0	1.0	582.0	1.0	20.0	0.0	260012.0	1.0	122.0	1.0
4	53.0	1.0	81.0	1.0	60.0	0.0	263550.0	1.0	138.0	1.0
5	86.5699573508972 3	1.0	1655.0	0.0	25.0	0.0	266000.0	1.9	137.0	1.0
6	50.0	0.0	61.0	1.0	25.0	0.0	135013.0	1.2	136.0	1.0
7	60.0	0.0	2656.0	1.0	30.0	0.0	305000.0	2.3	137.0	1.0
8	45.0	1.0	130.0	0.0	35.0	0.0	174000.0	0.8	139.0	1.0
9	45.0	0.0	582.0	1.0	55.0	0.0	543000.0	1.0	132.0	0.0
10	65.0	1.0	59.0	1.0	60.0	0.0	172000.0	0.9	137.0	0.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	65.0	0.0	148.0	0.0	40.0	0.0	185000.0	1.3	135.0	1.0
2	62.0	1.0	91.0	0.0	43.0	1.0	201910.0	1.0	133.0	1.0
3	70.0	1.0	167.0	0.0	25.0	0.0	234345.0	1.0	134.0	1.0
4	42.0	0.0	133.0	0.0	25.0	0.0	155132.0	0.9	140.0	1.0
5	65.0	1.0	178.0	1.0	47.0	1.0	192266.0	1.18	136.0	0.0
6	70.0	1.0	109.0	1.0	35.0	1.0	263358.0	1.2	135.0	1.0
7	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0
8	79.0	1.0	55.0	0.0	50.0	1.0	172000.0	1.8	133.0	1.0
9	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
10	70.0	0.0	92.0	0.0	60.0	1.0	317000.0	0.8	140.0	0.0

TabuLLM: AI Benchmark Report

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	70.0	1.0	318.0	0.0	50.0	0.0	254355.0	1.68243985191822 4	140.0	0.0
2	71.9250023109793 2	0.0	969.0	1.0	38.0	0.0	331648.0	1.83	130.0	0.0
3	60.0	0.0	925.0	1.0	45.0	0.0	203000.0	3.00382413920101 5	132.0	0.0
4	49.0	0.0	96.0	0.0	30.0	1.0	223474.0	0.9	136.0	0.0
5	63.0	0.0	582.0	0.0	40.0	0.0	263175.0	2.12654789103866 9	140.0	1.0
6	85.0	1.0	132.0	1.0	20.0	0.0	220000.0	1.7	135.0	1.0
7	48.0	1.0	131.0	1.0	30.0	1.0	244000.0	1.6	130.0	0.0
8	95.0	1.0	371.0	0.0	30.0	0.0	461000.0	2.0	132.0	1.0
9	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
10	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	60.9042072833447 46	0.0	902.0	0.0	35.0	1.0	227671.0	0.8	144.0	1.0
2	50.0	1.0	64.0	0.0	50.0	1.0	231000.0	0.9	137.0	0.0
3	80.0	1.0	76.0	0.0	38.0	0.0	151460.0	1.7	143.0	1.0
4	65.1387575186934 4	1.0	219.0	1.0	40.0	0.0	243382.0	1.1	125.0	1.0
5	60.0	0.0	1691.0	0.0	40.0	0.0	259616.0	0.8	139.0	0.0
6	60.0	0.0	134.0	1.0	30.0	1.0	271000.0	1.83	130.0	0.0
7	78.0	0.0	224.0	0.0	50.0	0.0	481000.0	1.4	138.0	1.0
8	70.0	0.0	582.0	0.0	20.0	1.0	263358.03	1.83	134.0	1.0
9	61.0	0.0	248.0	0.0	30.0	1.0	267000.0	0.7	136.0	1.0
10	40.0	0.0	478.0	1.0	30.0	0.0	303000.0	0.9	136.0	1.0

TabuLLM: AI Benchmark Report

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	55.0	1.0	582.0	1.0	35.0	1.0	449636.0	1.0	135.0	0.0
2	60.0	0.0	2436.0	0.0	30.0	0.0	265000.0	1.0	140.0	1.0
3	65.0	0.0	605.0	0.0	38.0	1.0	350299.0	0.9	138.0	1.0
4	51.0	0.0	1255.0	1.0	40.0	0.0	301916.0	0.6481396065861863	147.0	1.0
5	40.0	0.0	92.0	1.0	40.0	1.0	173000.0	1.1	139.0	1.0
6	70.0	0.0	370.0	1.0	61.0	0.0	226000.0	1.3	136.0	1.0
7	81.0	0.0	4540.0	0.0	35.0	0.0	231000.0	1.18	137.0	1.0
8	70.0	0.0	2695.0	1.0	40.0	0.0	241000.0	1.0	137.0	1.0
9	70.0	0.0	835.0	0.0	35.0	1.0	305000.0	0.8	133.0	0.0
10	80.0	0.0	776.0	1.0	38.0	1.0	192000.0	1.3	135.0	0.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	60.0	0.0	2691.0	1.0	40.0	1.0	218250.0	0.9	137.0	1.0
2	45.0	0.0	582.0	0.0	35.0	0.0	420164.0	1.0	138.0	1.0
3	52.0	0.0	834.0	0.0	40.0	0.0	302000.0	0.8	138.0	1.0
4	50.637363290827366	1.0	250.0	0.0	25.0	0.0	175118.0	1.2	137.0	0.0
5	69.0	1.0	65.0	1.0	45.0	0.0	225642.0	1.0	134.0	0.0
6	69.0	1.0	129.0	1.0	38.0	0.0	63913.0	1.18	141.0	0.0
7	75.0	0.0	582.0	1.0	30.0	1.0	263358.03	1.83	134.0	0.0
8	53.0	0.0	207.0	1.0	40.0	0.0	223000.0	1.2	130.0	0.0
9	60.0	1.0	2281.0	1.0	40.0	0.0	283000.0	1.0	141.0	0.0
10	59.0	1.0	280.0	1.0	25.0	1.0	302000.0	1.0	141.0	0.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	45.0	0.0	582.0	1.0	20.0	0.0	327935.0	0.8	139.0	0.0
2	49.0	0.0	95.0	0.0	55.0	0.0	147000.0	0.9	139.0	1.0
3	45.0	0.0	167.0	0.0	30.0	0.0	328316.0	1.1	139.0	1.0
4	42.0	0.0	69.0	1.0	25.0	1.0	222164.0	0.7	140.0	1.0
5	69.0	0.0	2876.0	1.0	30.0	0.0	306989.0	2.4720287418898947	132.0	1.0
6	80.0	1.0	247.0	0.0	25.0	1.0	173522.0	2.0338194901313584	137.0	1.0
7	65.0	0.0	224.0	1.0	50.0	0.0	149000.0	1.3	137.0	1.0
8	80.0	0.0	898.0	0.0	25.0	0.0	149000.0	1.1	144.0	1.0
9	66.0	1.0	72.0	0.0	40.0	1.0	242000.0	1.2	134.0	1.0
10	75.0	1.0	582.0	0.0	30.0	0.0	225000.0	1.83	134.0	1.0

TabuLLM: AI Benchmark Report

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	58.0	0.0	151.0	0.0	35.0	0.0	166000.0	1.9573433829531866	136.0	1.0
2	65.0	0.0	56.0	1.0	60.0	0.0	153000.0	0.9	144.0	1.0
3	50.0	1.0	582.0	0.0	35.0	0.0	134218.0	1.3	134.0	1.0
4	44.0	0.0	5849.0	1.0	30.0	0.0	389432.0	0.8	138.0	0.0
5	45.0	0.0	95.0	0.0	40.0	0.0	184165.0	0.8	136.0	1.0
6	76.54620702061675	0.0	150.0	1.0	40.0	1.0	362941.0	6.036665197648335	136.0	1.0
7	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0
8	60.0	0.0	68.0	0.0	20.0	0.0	119000.0	2.9	127.0	1.0
9	70.0	0.0	97.0	0.0	60.0	1.0	220000.0	0.9	138.0	1.0
10	53.0	1.0	1808.0	0.0	60.0	1.0	249000.0	0.7	138.0	1.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	77.44983267919275	1.0	185.0	1.0	30.0	0.0	284556.0	1.4	135.0	0.0
2	65.0	1.0	167.0	1.0	35.0	0.0	244000.0	1.3	132.0	1.0
3	85.0	1.0	582.0	0.0	35.0	0.0	305915.0	1.7	136.0	1.0
4	70.0	1.0	64.0	1.0	35.0	1.0	216006.0	1.1	141.0	1.0
5	53.0	0.0	273.0	1.0	38.0	1.0	221000.0	1.0	139.0	0.0
6	60.0	0.0	620.0	0.0	40.0	1.0	277000.0	1.1	140.0	0.0
7	60.0	0.0	897.0	1.0	45.0	0.0	297000.0	1.0	133.0	1.0
8	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
9	45.0	1.0	1876.0	1.0	35.0	0.0	226000.0	0.9	138.0	1.0
10	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0